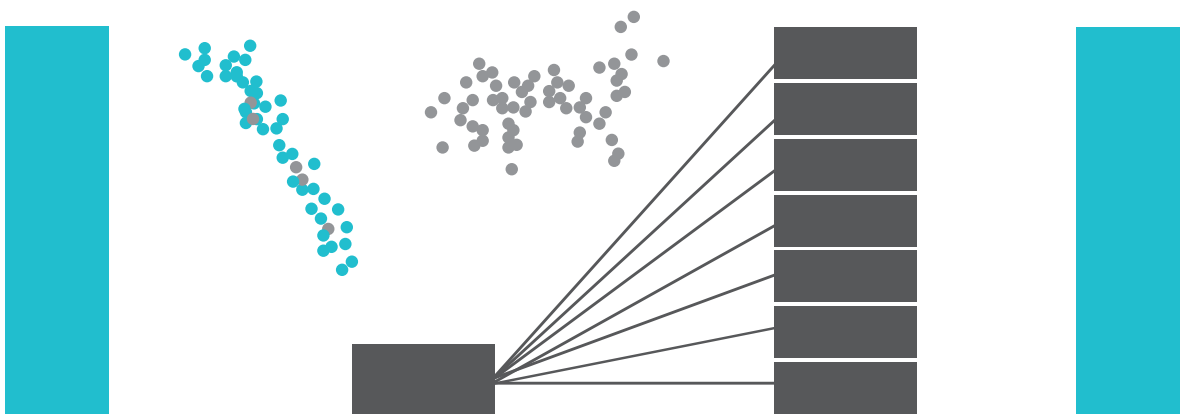


Upsampling and Downsampling: Correcting the Imbalances in Data

By mastering the art of upsampling and downsampling, we not only correct skewed datasets but also pave the way for fairer, more accurate machine learning models. This first article in the two-part series focuses on the uses of upsampling.



We rely on history for future predictions, but in reality, historical data may not be sufficiently representative for accurate forecasting. As an example, historically, most top-level executive positions have been occupied by men. However, it is crucial to ensure that women who are qualified and come from diverse backgrounds are also afforded equal opportunities. Another illustration of imbalance lies in loan applications, where approximately 90% are typically rejected and only 10% are approved. This skewed ratio can lead to erroneous decisions, resulting in good loan applications being unfairly rejected. Utilising such imbalanced data for model learning can yield biased outputs supporting the majority class in predictions. Numerous real-world scenarios exhibit such data imbalances.

Another factor contributing to biased outcomes is the method of data collection. Sampling, a data collection technique involving a small subset of the population, often introduces imbalance, yielding biased data.

To counter such imbalanced datasets, we use techniques called upsampling and downsampling.

Upsampling

Upsampling involves augmenting the number of instances in a dataset and is particularly useful when there is an imbalanced distribution of data, with one class significantly underrepresented. This process involves generating synthetic data points using different techniques or generative models akin to the existing data and integrating them into the dataset. However, it is important to guard against overfitting by ensuring that the addition of new data doesn't adversely affect the model. Following this process, the count of the classes is almost balanced.

Methods used in upsampling

Upsampling is applicable in scenarios where the dataset exhibits an imbalanced distribution or is relatively small.